

ELBOW, SK, Canada

WMO: 714500

Lat: 51.1256N

Lon: 106.5826W

Elev: 595

StdP: 94.38

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-29.1	-35.3	0.1	-31.8	-32.2	0.2	-29.0	13.8	-9.9	12.3	-10.7	3.2	290	0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	31.3	18.2	29.0	17.6	27.0	16.7	20.6	27.4	19.3	26.0	18.2	24.6	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	14.2	23.6	17.0	13.0	21.9	15.8	12.0	20.8	62.1	27.3	57.6	26.0	53.8	24.5	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.5	9.4	-35.0	35.0	2.3	2.9	-36.6	37.1	-38.0	38.8	-39.3	40.5	-40.9	42.6
			-35.0	23.0	2.2	1.3	-36.6	23.9	-37.8	24.7	-39.1	25.5	-40.7	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-13.9	-12.0	-4.8	4.2	10.9	15.7	18.7	17.9	12.9	4.9	-3.9	-11.2	(6)
(7)		DBStd	13.10	8.68	7.97	8.08	5.57	4.57	3.43	3.04	3.52	4.52	5.02	6.83	8.11	(7)
(8)		HDD10.0	3312	740	617	458	183	45	2	0	1	22	169	417	658	(8)
(9)		HDD18.3	5567	998	851	716	423	234	91	32	51	171	416	667	916	(9)
(10)		CDD10.0	889	0	0	0	9	72	174	271	244	108	11	0	0	(10)
(11)		CDD18.3	103	0	0	0	0	2	13	44	37	7	0	0	0	(11)
(12)		CDH23.3	1579	0	0	0	6	83	200	557	558	171	4	0	0	(12)
(13)		CDH26.7	508	0	0	0	1	16	49	184	206	53	0	0	0	(13)
(14)	Wind	WSAvg	4.6	4.6	4.4	4.9	5.1	5.1	4.5	3.9	4.0	4.4	4.7	4.7	4.4	(14)
(15)	Precipitation	PrecAvg	378	17	12	14	28	46	76	54	43	31	20	13	16	(15)
(16)		PrecMax	566	32	32	39	84	118	153	124	106	81	68	40	40	(16)
(17)		PrecMin	152	2	5	1	6	0	11	9	3	2	1	0	5	(17)
(18)		PrecStd	90	7	6	10	20	34	37	31	26	22	16	11	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.8	14.3	23.1	28.9	31.2	34.1	35.1	31.2	23.3	15.1	7.1	(19)
(20)			MCWB	2.6	2.6	7.7	11.7	14.2	17.4	20.8	19.4	16.4	13.2	8.5	3.4	(20)
(21)		2%	DB	3.1	2.9	10.4	19.7	25.6	27.7	30.9	31.6	28.0	19.0	9.9	4.2	(21)
(22)			MCWB	1.1	0.8	5.6	9.9	13.4	17.0	19.4	18.0	15.3	10.7	5.6	1.6	(22)
(23)		5%	DB	1.1	1.1	7.4	16.7	23.1	25.5	28.6	28.8	24.9	15.6	6.9	2.1	(23)
(24)			MCWB	-0.3	-0.3	3.9	8.4	12.1	15.7	18.8	17.3	14.3	9.1	3.7	0.4	(24)
(25)		10%	DB	-1.5	-0.7	4.6	13.7	20.4	23.4	26.4	26.4	21.8	12.8	4.4	-0.2	(25)
(26)			MCWB	-2.6	-1.7	2.2	7.1	10.9	14.8	17.8	16.6	12.8	7.7	2.2	-1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.5	8.3	12.7	16.3	21.0	23.5	21.2	18.1	13.7	8.9	3.5	(27)
(28)			MCDWB	5.2	4.6	13.4	20.0	24.6	27.4	30.3	29.7	27.9	21.9	15.1	6.7	(28)
(29)		2%	WB	1.2	0.8	5.8	10.6	14.5	18.7	21.2	19.7	16.2	11.5	6.0	1.7	(29)
(30)			MCDWB	2.9	2.2	10.0	17.9	22.9	24.9	27.5	27.3	25.1	17.6	9.8	3.8	(30)
(31)		5%	WB	-0.2	-0.4	3.9	9.0	13.1	17.1	19.9	18.7	15.0	9.7	4.1	0.2	(31)
(32)			MCDWB	1.4	0.7	7.2	16.0	21.3	22.8	26.3	26.0	23.2	14.3	6.8	1.8	(32)
(33)		10%	WB	-2.2	-2.0	2.2	7.4	11.9	15.9	18.7	17.6	13.6	8.2	2.3	-1.6	(33)
(34)			MCDWB	-1.1	-1.0	4.2	12.8	19.4	21.8	24.9	24.4	20.6	12.5	4.2	-0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.1	9.9	12.1	13.7	11.8	13.0	13.8	13.5	11.3	9.0	9.0	(35)
(36)			MCDBR	11.0	10.8	12.4	17.8	18.1	15.8	16.9	18.8	19.7	16.5	12.8	10.8	(36)
(37)			MCWBR	9.4	9.4	8.4	9.2	8.3	7.2	7.7	7.5	8.7	9.3	8.5	8.5	(37)
(38)		5% WB	MCDWR	11.2	10.7	12.4	16.3	16.1	13.1	15.0	15.4	17.2	15.4	12.3	10.6	(38)
(39)			MCWBR	9.7	9.2	8.4	9.0	7.6	6.8	7.7	7.2	8.1	9.0	8.4	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.254	0.270	0.334	0.350	0.355	0.365	0.370	0.325	0.282	0.252	0.223	(40)	
(41)		taud	2.304	2.363	2.488	2.369	2.372	2.409	2.391	2.356	2.460	2.559	2.493	2.410	(41)	
(42)		Ebn at Noon	778	897	948	907	904	900	884	859	868	858	801	789	(42)	
(43)		Edh at Noon	64	79	85	109	114	111	112	110	88	66	53	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.15	2.15	3.53	4.79	5.86	6.03	6.51	5.23	3.83	2.30	1.20	0.85	(44)	
(45)		RadStd	0.12	0.16	0.28	0.41	0.43	0.44	0.38	0.38	0.35	0.30	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.03	-1.26	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page